

Smart Attendance System - Status Update

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What's built

- Android app (Kotlin, Jetpack Compose, MVVM, Hilt, Room, CameraX) - professor login, dashboard, live camera attendance, student enrollment, history, analytics, PDF/Excel export, WhatsApp low-attendance alerts.
- Face recognition pipeline using InsightFace models (SCRFD detector + ArcFace embedder) running on-device via ONNX Runtime - detect, align, embed, match, all offline.
- Backend: Spring Boot + PostgreSQL, JWT-secured REST API, offline-first sync from the app.
- Full documentation set: architecture, database schema, install/deploy/testing guides.

Bugs found and fixed this session

- App crashed right after login: the face-recognition engine was loading model files eagerly as soon as the Dashboard screen was built, even before the camera was ever opened. Fixed by loading models lazily, only when actually needed.
- App crashed when capturing a photo during student enrollment: the capture used the wrong image format handling (mixed up two different camera output formats). Fixed with the correct decode path for that capture mode.
- Face detector was silently finding zero faces: the downloaded model's internal output names were plain numbers, not the descriptive names the code expected. Fixed by identifying each output by its data shape instead of by name.
- 'Start Attendance' looked disabled: the dashboard defaulted to a semester that had no demo subjects configured. Fixed the default, and added a way to add new subjects directly from the app (Department/Electronics/Python/etc.).
- Attendance always showed 'Present: 0/0': a newly-enrolled student was silently saved under different Department/Semester/Section defaults than the one selected for taking attendance, so there was nobody to match against. Fixed by making the enrollment form always default to whatever is currently selected on the Dashboard.
- Backend startup failure: a database column type did not match what the backend code expected. Fixed the schema definition.
- Various local environment issues resolved: Docker networking, Windows Firewall rule for the phone to reach the backend over Wi-Fi, and the PC's Wi-Fi IP address changing overnight (DHCP), which needed the app's backend address updated.

Current status

Login, Dashboard, adding subjects, student enrollment (face capture), Start Attendance camera recognition, History, Analytics, and PDF/Excel export are all working end to end on a real device, talking to a backend running on the development PC over local Wi-Fi.

Known limitations / next steps

- The backend currently runs on the development PC, reachable only over the same Wi-Fi network - not yet deployed anywhere permanent/public.
- The app is a debug build, not signed for release - fine for testing, not yet ready for Play Store distribution.
- Recognition accuracy/threshold has not yet been tuned with real classroom testing across multiple students, lighting conditions, etc.
- WhatsApp low-attendance alerts currently open WhatsApp with a prefilled message for the professor to review and send manually (no automatic sending).